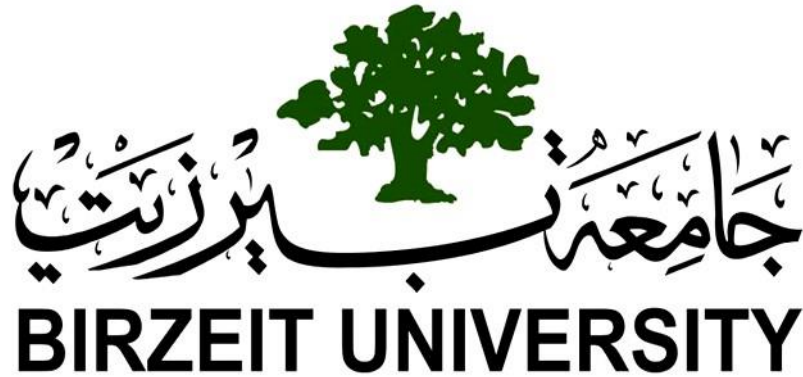




Faculty of Engineering & Technology

Electrical & Computer Engineering Department

CIRCUITS AND ELECTRONICS LABORATORY ENEE2103

Report#3

Experiment 5: Filters

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Abstract

In this experiment, different filters are examined, including passive filters of first and second order, as well as active filters. The purpose of the experiments is to study the characteristics of first-order configurations, such as high-pass and low pass filters, and second-order configurations, such as band-pass and band-stop filters. Additionally, the study investigates active low-pass filters. The objective is to present and demonstrate the principles and capabilities of these filters across different signal processing scenarios.

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Theory

First Order Passive Filters

we can create passive first order filters efficiently and affordably by simply connecting one resistor and one capacitor in a series configuration to the input signal (V_{IN}). The location of these elements determines the V_{OUT} of the filter.

If a resistor and capacitor are placed in a position relative to the output signal, then a resistor and capacitor are considered to be positioned correctly then a resistor and capacitor are considered to be positioned correctly. It is best to use either a High Pass Filter or a low Pass Filter [1].

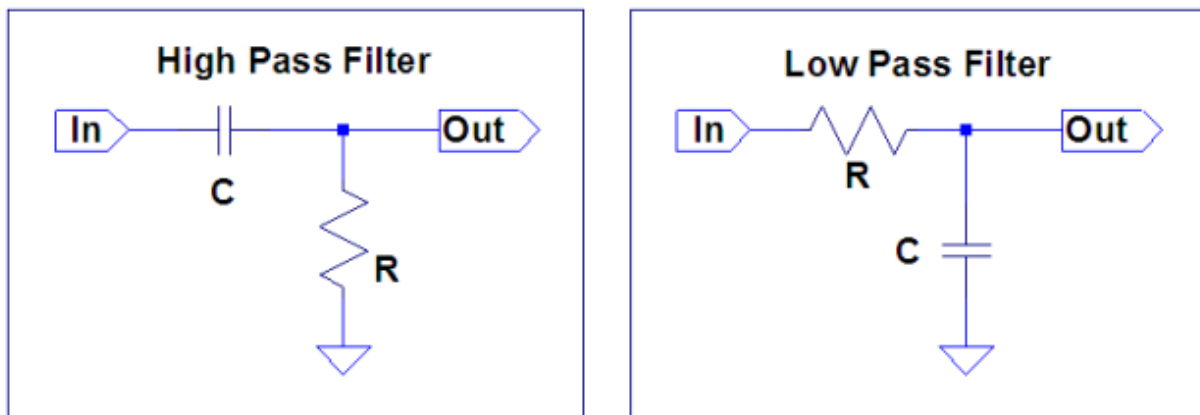


Figure 1: type of filter according to circuit elements order (R, C)

Filters work by reducing the overall amplitudes of a signal's frequencies, while enabling a specific range of frequencies to pass through without any changes. The theoretical frequency response curves of the 4 basic filter types determine the desired performance of any filter [1].

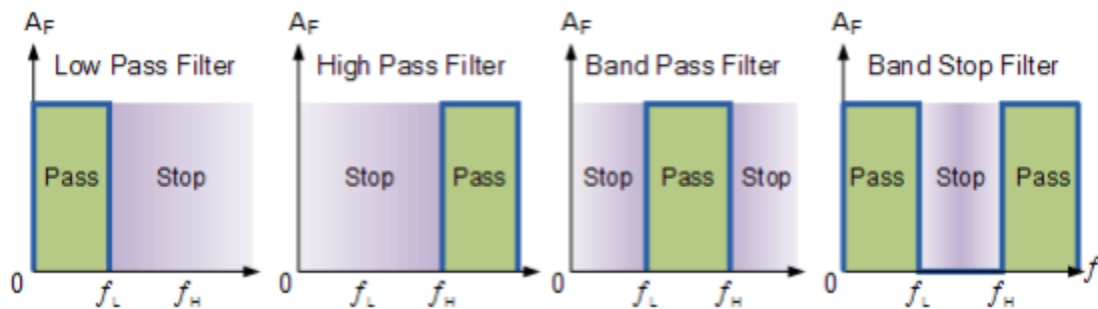


Figure 2: how each filter behaves according to frequency range

Second Order Passive Filters

An RLC circuit used in electronics usually consists of three elements: a resistor (R), an inductor (L) along with a capacitor (C). Each of them is crucial to change the circuit behavior, especially in filtering applications. The resistor acts as current regulator to maintain a safe flow of electricity. It does this by creating a negative voltage drop on current. The inductor stores energy in a magnetic field. Its resistance to abrupt changes in current flow makes it indispensable for smoothing rapid current fluctuations - a crucial property of effective filtering [2].

The combination of capacitors storing energy in electric field and permitting alternating current (AC) while blocking direct current (DC) makes this capacitor an important component of voltage and current control. This is necessary in filter circuits where capacitors remove unwanted frequencies or increase desired ones. When combined in an RLC circuit, they may be configured as efficient filters that alter the frequency and quality of electrical signals traveling through them [2].

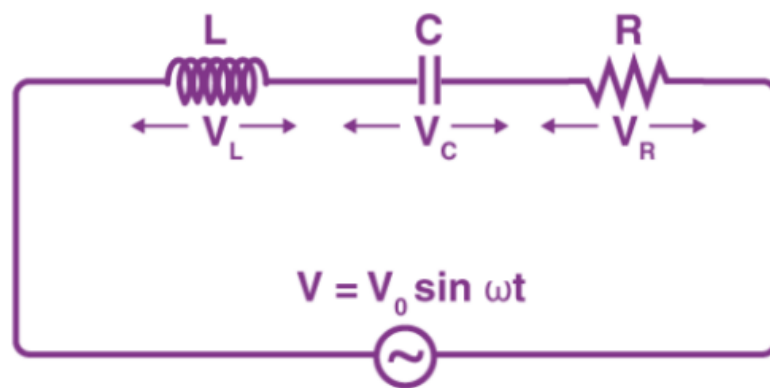


Figure 3: Second order RLC filter

Active Filters

Active filters (especially those used for low-pass filtering) employ operational amplifiers (op-amps) along with resistors and capacitors to amplify signals. These filters are good helpers for low frequency signals. Unlike passive filters which weaken the signal strength by "Attenuation," active filters add power back into the signal.

This power boost is obtained by active filters using operational amplifiers, Transistors or FETs (Field-Effect transistors). These parts need outside power to work. They take this power and use it to strengthen or amplify the signal to make the filter job easier. So, active filters are useful

when you want to tump down a signal without it losing strength or power [3].

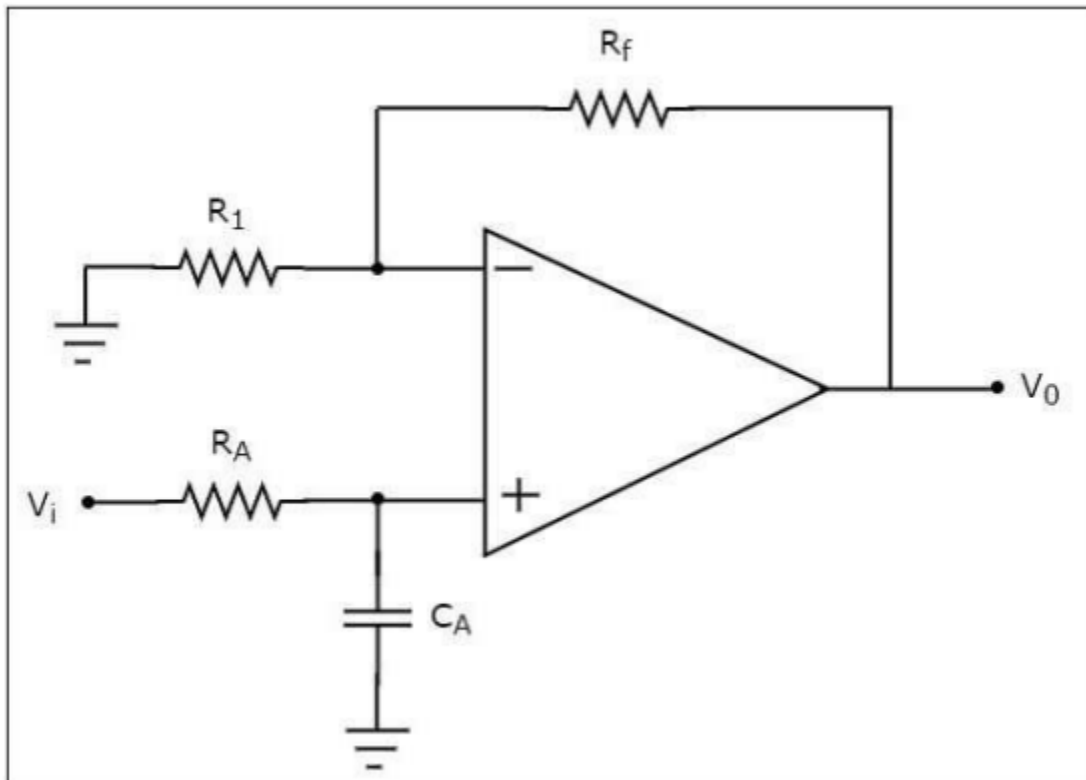


Figure 4: Active Filter using Op-Amp

Procedure, Calculations and Discussions

RC Circuit

First order filters

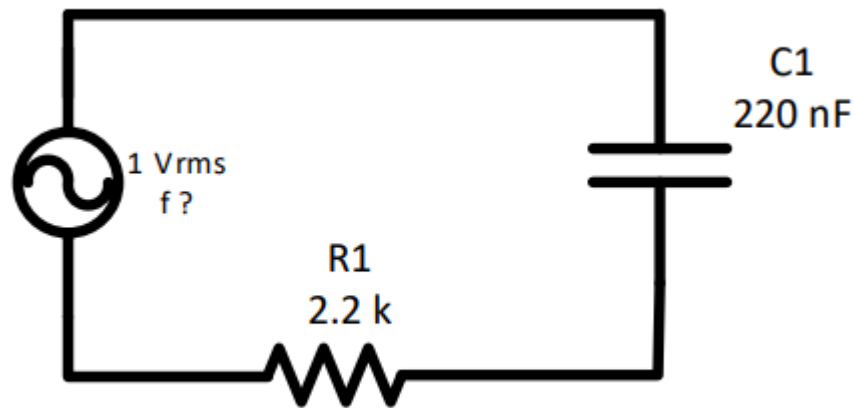


Fig (5.1)

Figure 5: First order circuit

SET		MEASURE		CALCULATE		
Frequency(Hz)	Vin(Vrms)	VR(Vrms)	VC(Vrms)	log f	20 log (VR/Vin)	20 log (VC/Vin)
20	1	0.0685	1.07	1.30103	-23.28618857	0.587675554
32.8	1	0.104	0.99	1.515874	-19.65933321	-0.087296108
164	1	0.48	0.9	2.214844	-6.375175252	-0.915149811
328	1	0.7	0.7	2.515874	-3.0980392	-3.0980392
1312	1	0.97	0.23	3.117934	-0.264565315	-12.76544328
3280	1	0.99	0.096	3.515874	-0.087296108	-20.35457534
4592	1	0.99	0.07	3.662002	-0.087296108	-23.0980392
6560	1	1	0.048	3.816904	0	-26.37517525

Table 1 : table 1 first order circuits:

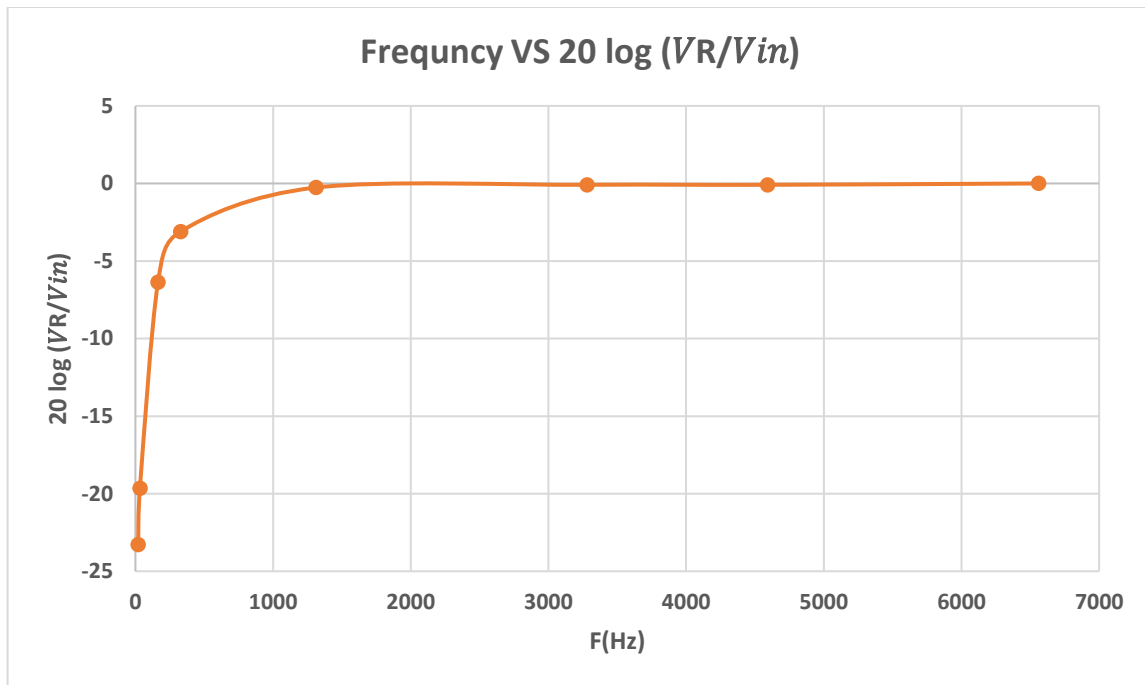


Figure 6: High pass filter

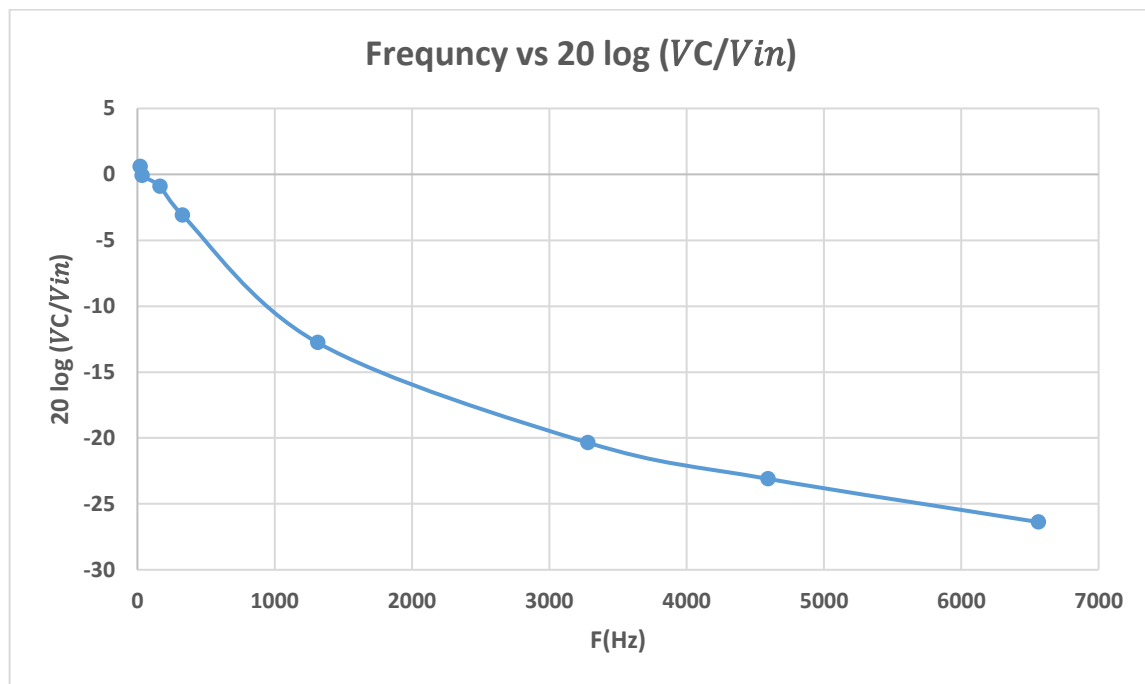


Figure 7:passive low pass filter

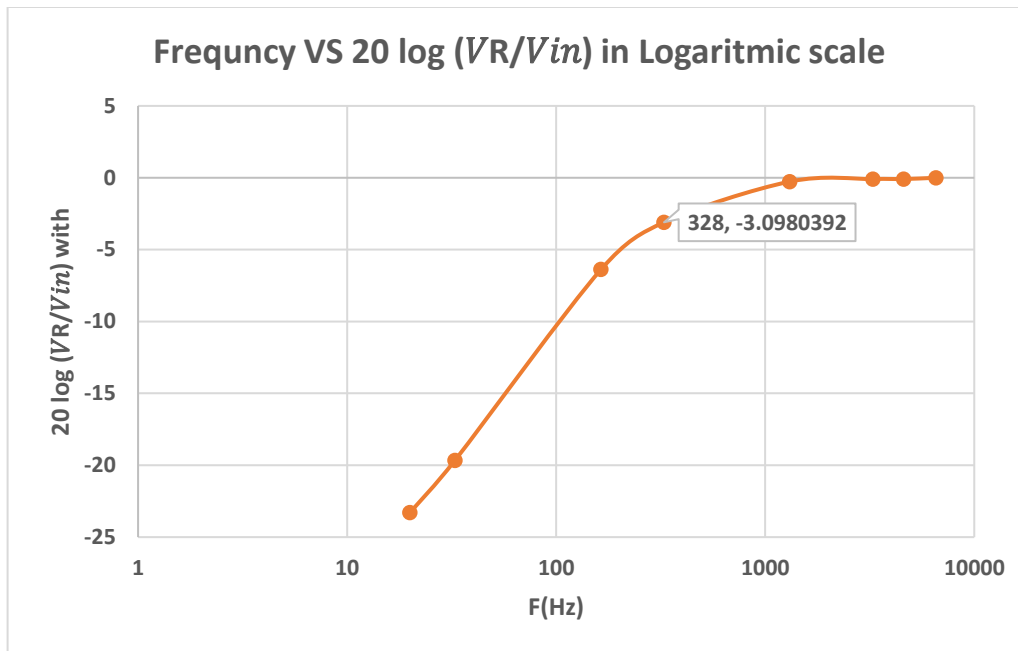


Figure 8: cut off frequency of low pass filter

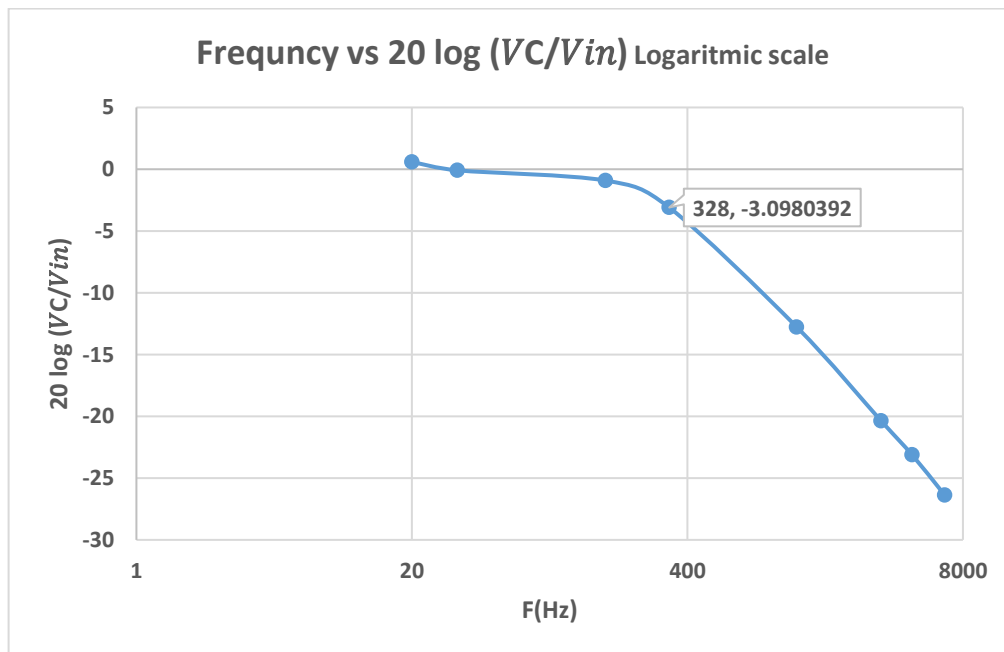


Figure 9: cut off frequency of high pass filter.

Filter Type Determination:

The behavior of the V_r / V_{in} and V_c / V_{in} ratios as frequency changes is essential to determine the filter type. The V_r/V_{in} begins very low at 20 Hz, indicating high attenuation at lower frequencies. The higher the frequency is, the higher the V_r/V_{in} rise which shows the filter is letting more of the higher frequencies to pass. Which this behaves like a high-pass filter.

On the other hand V_c / v_{in} starts higher at 20 Hz with low attenuation at low frequencies, but when we increase the frequency the ratio is decreased. This behavior shows a low-pass filter, which allows low frequencies to pass and attenuates higher frequencies.

Frequency:

$$\text{In theory: } f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi * 2.2k * 220n} = 328.8$$

The point at which the high-pass filter's cut off frequency is, when the decrease in $20\log(V_r / V_{in})$ reaches -3 dB. At a frequency of 328 Hz, the recording shows a value of -3.10 dB by using the formula $20\log(V_r / V_{in})$.

For the low-pass filter, the 3-dB cut-off frequency is also when $20\log(V_c/V_{in})$ decreasing to -3 dB. According to the table above at 328 Hz, $20\log(V_c/V_{in}) = -3.10$ dB. Which means in both cases the cut off frequency is very close to the theoretical value which is 328.

Phase Shift Analysis:

For a high-pass filter the phase shift would be from near +90 for low frequencies and to 0 for frequencies well above the cut-off frequency. In other hand the low-pass filter, this phase shift will be from zero to 90 at higher frequencies.

Second order filters

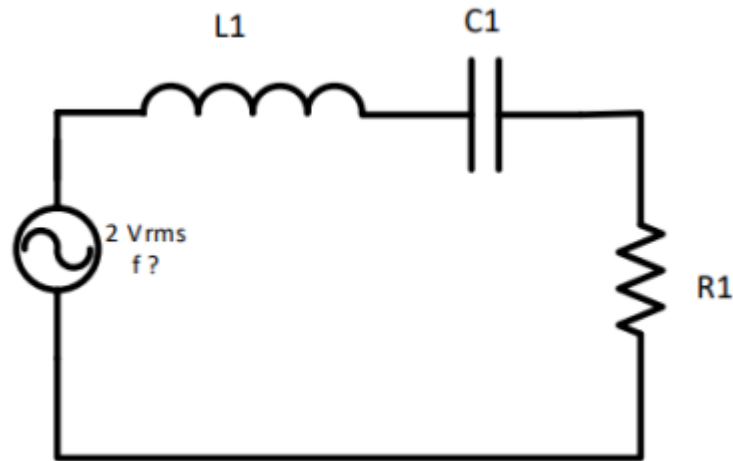


Figure 10: Second order filter in the manual

SET		MEASURE		CALCULATE		
Frequency(Hz)	Vin(Vrms)	VR(Vrms)	(VC+ VL)(Vrms)	log f	20 log (VR/Vin)	20 log (VC + VL /Vin)
26.5	2	0.178	2.105	1.423246	-21.01219987	0.44444209
265	2	1.37	1.52	2.423246	-3.28618857	-2.383728154
340	2	1.599	1.234	2.531479	-1.943630638	-4.194296719
680	2	1.98	0.226	2.832509	-0.087296108	-18.93843113
1360	2	1.73	1.059	3.133539	-1.259677851	-5.522680711
2000	2	1.39	1.51	3.30103	-3.160303908	-2.441060967
20000	2	0.041	2.108	4.30103	-33.76492278	0.456812218
28000	2	0.051	2.099	4.447158	-31.86919639	0.419648858
40000	2	0.147	2.077	4.60206	-22.67425322	0.328130017

Table 2: Second order filter table

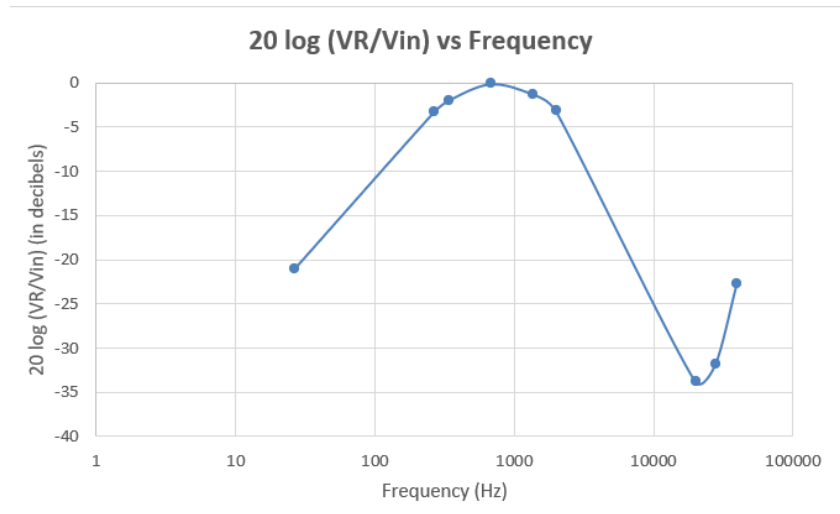


Figure 11: band-pass filter

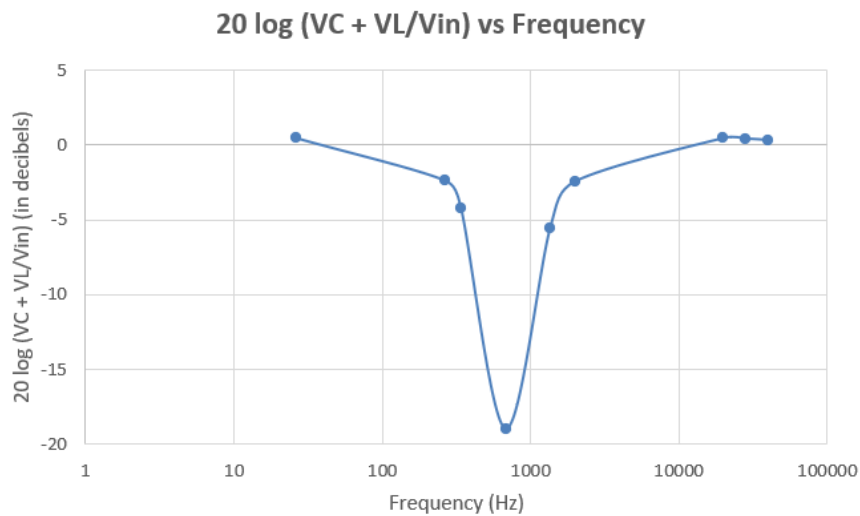


Figure 12: Band-stop (notch) filter

Figure 11 shows the VR/V_{in} ratio as the frequency increases up to a certain point and then decreases, forming a peak. This is characteristic of a band-pass filter that lets a certain band of frequencies to pass but attenuates frequencies outside this band.

Figure 12 displays the $(VC + VL)/V_{in}$ ratio with a sharp dip in the middle frequency range. This is typical of a band-stop (notch) filter that attenuates a particular band of frequencies and lets those outside the band pass without filter them.

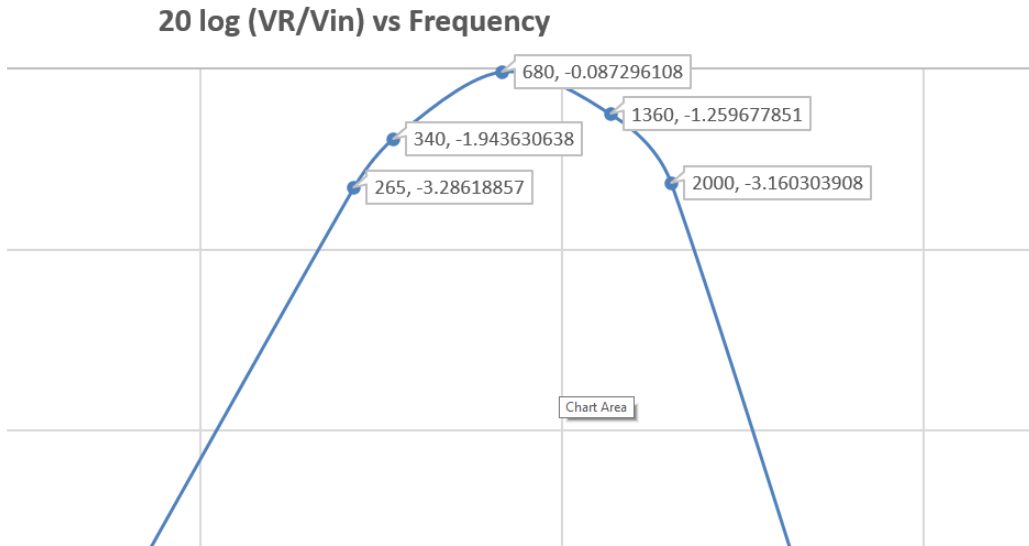


Figure 13: values of band pass

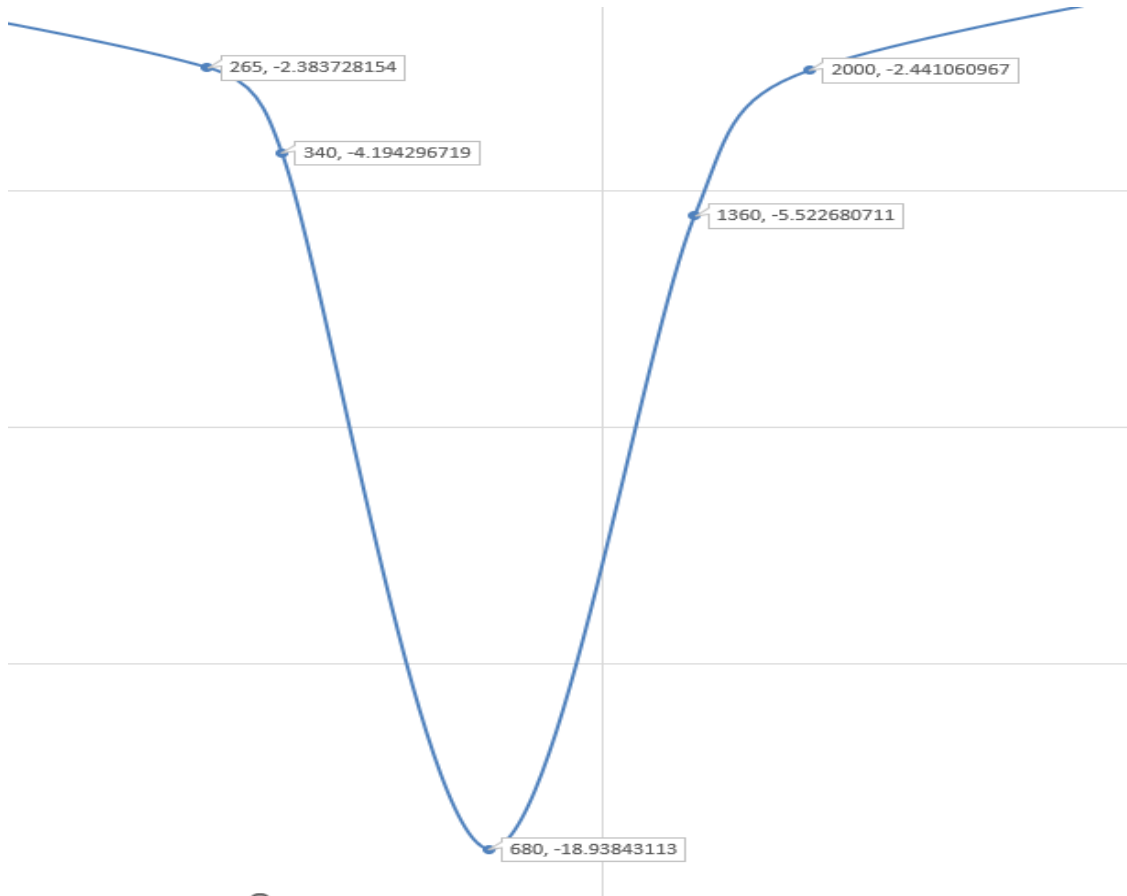


Figure 14: band stop values

We can see $f_{c1} = 265 \text{ Hz}$, $f_{c2} = 2000 \text{ Hz}$, $f_0 = 680 \text{ Hz}$, from the both figures above .

Theoretical values :

$$1. \text{The resonance frequency } f_0 = \frac{1}{2\pi\sqrt{L_1 C_1}} = \frac{1}{2\pi\sqrt{100 \times 10^{-3} \cdot 470 \times 10^{-9}}} = 734.1 \text{ Hz}$$

$$\text{The lower cut off frequency } = f_{c1} = \frac{1}{2\pi} \left(\sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} - \frac{R}{2L} \right)$$

$$f_{c1} = \frac{1}{2\pi} \left(\sqrt{\left(\frac{1 \times 10^3}{2 \cdot 100 \times 10^{-3}}\right)^2 + \frac{1}{100 \times 10^{-3} \cdot 470 \times 10^{-9}}} - \frac{1 \times 10^3}{2 \cdot 100 \times 10^{-3}} \right) = 287 \text{ Hz}$$

3. The upper cut-off frequency :

$$f_{c2} = \frac{1}{2\pi} \left(\sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} + \frac{R}{2L} \right)$$

$$f_{c2} = \frac{1}{2\pi} \left(\sqrt{\left(\frac{1 \times 10^3}{2 \cdot 100 \times 10^{-3}}\right)^2 + \frac{1}{100 \times 10^{-3} \cdot 470 \times 10^{-9}}} + \frac{1 \times 10^3}{2 \cdot 100 \times 10^{-3}} \right) = 1879 \text{ Hz}$$

As we see, the practical values are near the theoretical ones. So, the experiment was successful.

There are some error due to some problems in the component.

Active filters

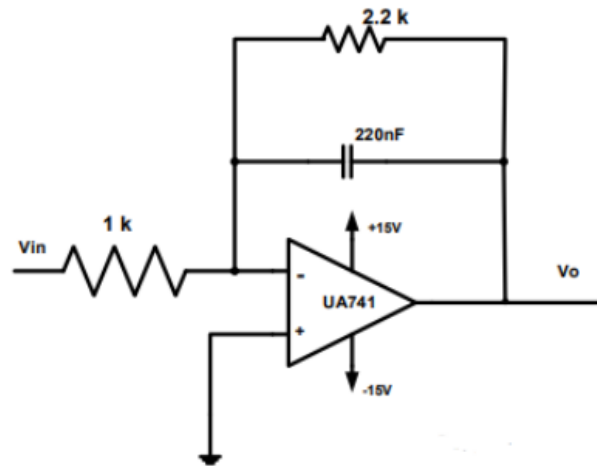


Figure 15: Active filter in the manual

This non-inverting active low pass filter , we can get F_c by $= 1/(2 \pi R_c)$

- $R=2.2 \text{ k}\Omega$
- $C=220 \text{ nF}$

$$F_c = 328.83 \text{ Hz}$$

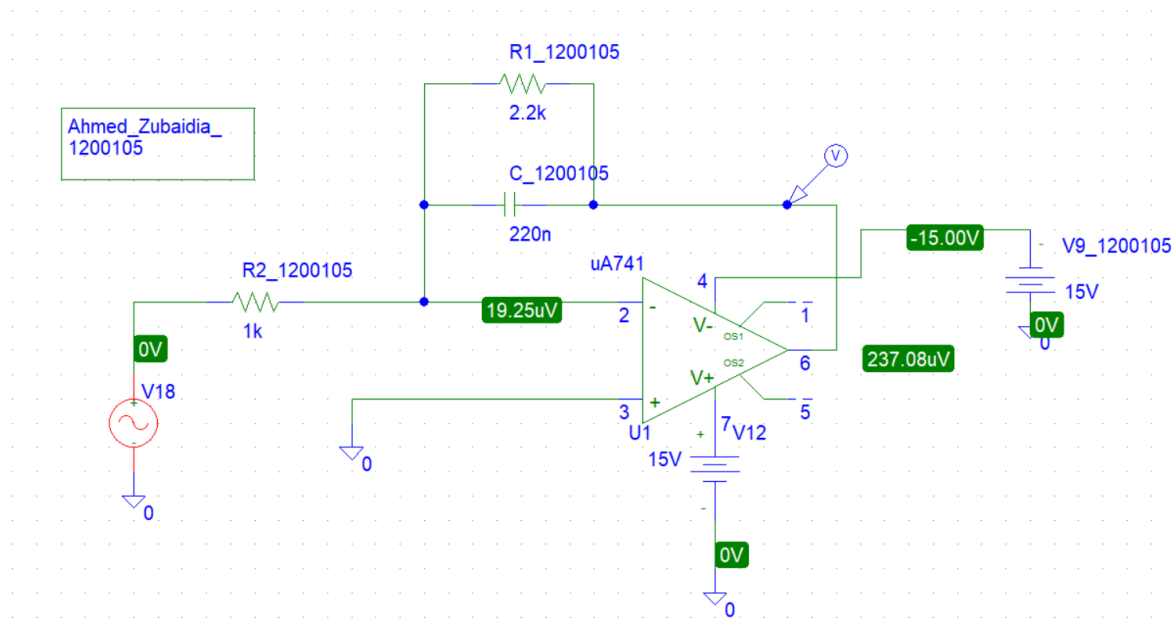


Figure 16: circuit Pspice

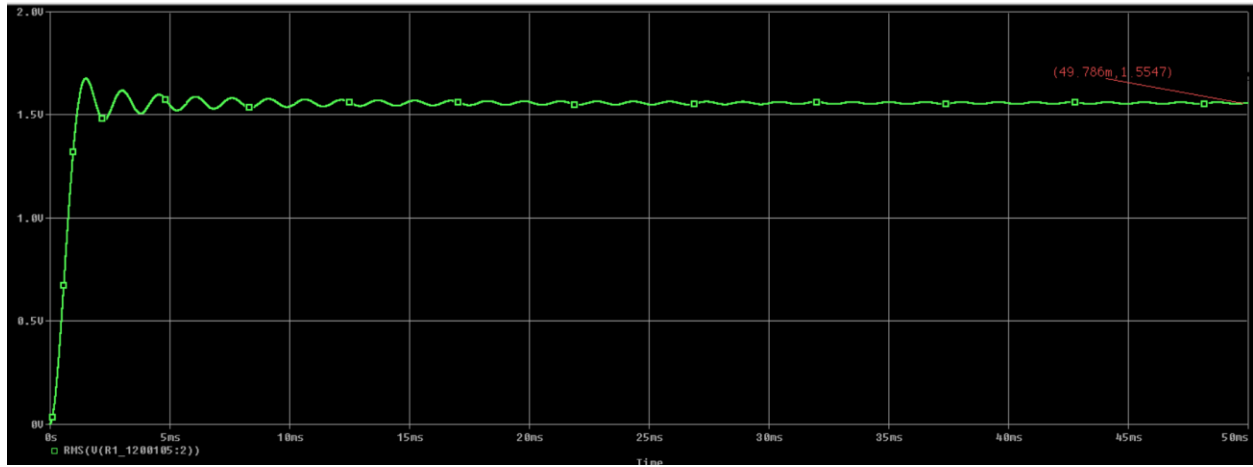


Figure 17: V_{out} rms at $f = 263.064$ Hz

SET	MEASURE		CALCULATE	
Frequency(Hz)	$V_{in}(V_{rms})$	$V_{out}(V_{rms})$	$\log f$	$20 \log (V_{out}/V_{in})$
32.883	1	2.2124	1.516971432	6.897272996
131.532	1	2.0334	2.119031424	6.164456384
263.064	1	1.766	2.420061419	4.939813985
328.38	1	1.5547	2.516376699	3.832931971
493.245	1	1.224	2.693062692	1.755628356
1315.32	1	0.535	3.119031424	-5.43292436
3288.3	1	0.219	3.516971432	-13.1911177
6576.6	1	0.11	3.818001428	-19.1721463

Table 3: noninverting active low pass filter table

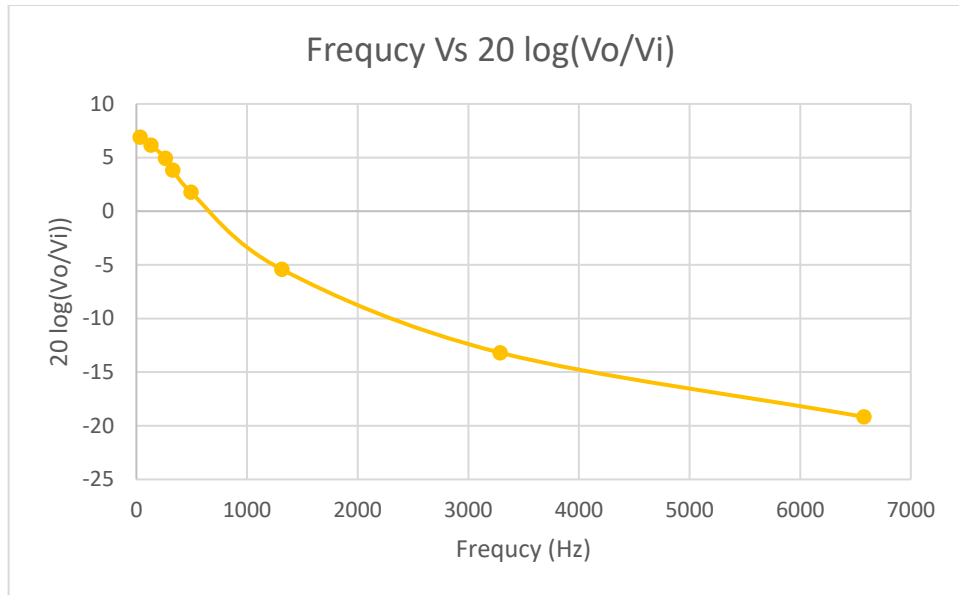


Figure 18: $(20 \log(V_o/V_i))$ vs frequency

As the frequency increases the output voltage is decreasing to zero, so this attitude tends to be a low pass filter. And if we look to the table we will notice that 328 hz is cut off frequency according to 0.7 max value which = 1.547 and the frequency at this voltage is 328 hz.

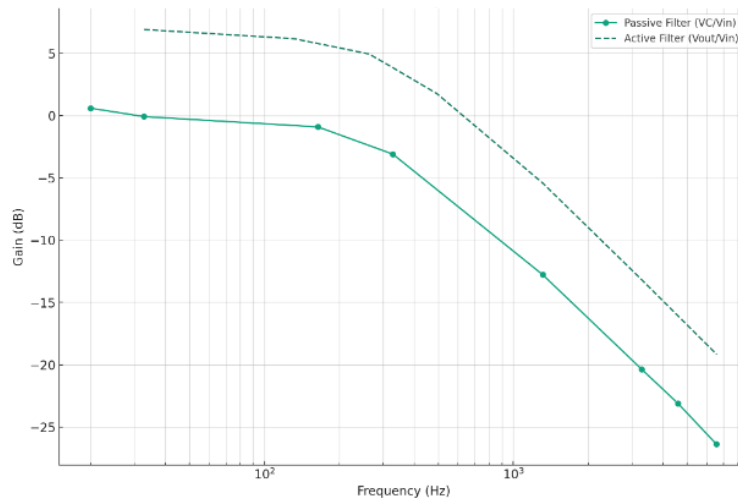


Figure 19: compare of passive and active filter

The active filter first amplifies low-frequency signals after which it starts reducing the signal strength with increasing frequency. The passive filter, however, maintains a constant signal strength for low-frequency signals and starts to decrease the signal strength without amplification. The active filter slowly reduces the power of high frequency signals, so that additional frequencies may be passed through it before it begins to decrease the signal strength. This makes the active filter suitable for situations where you need to both amplify signals and

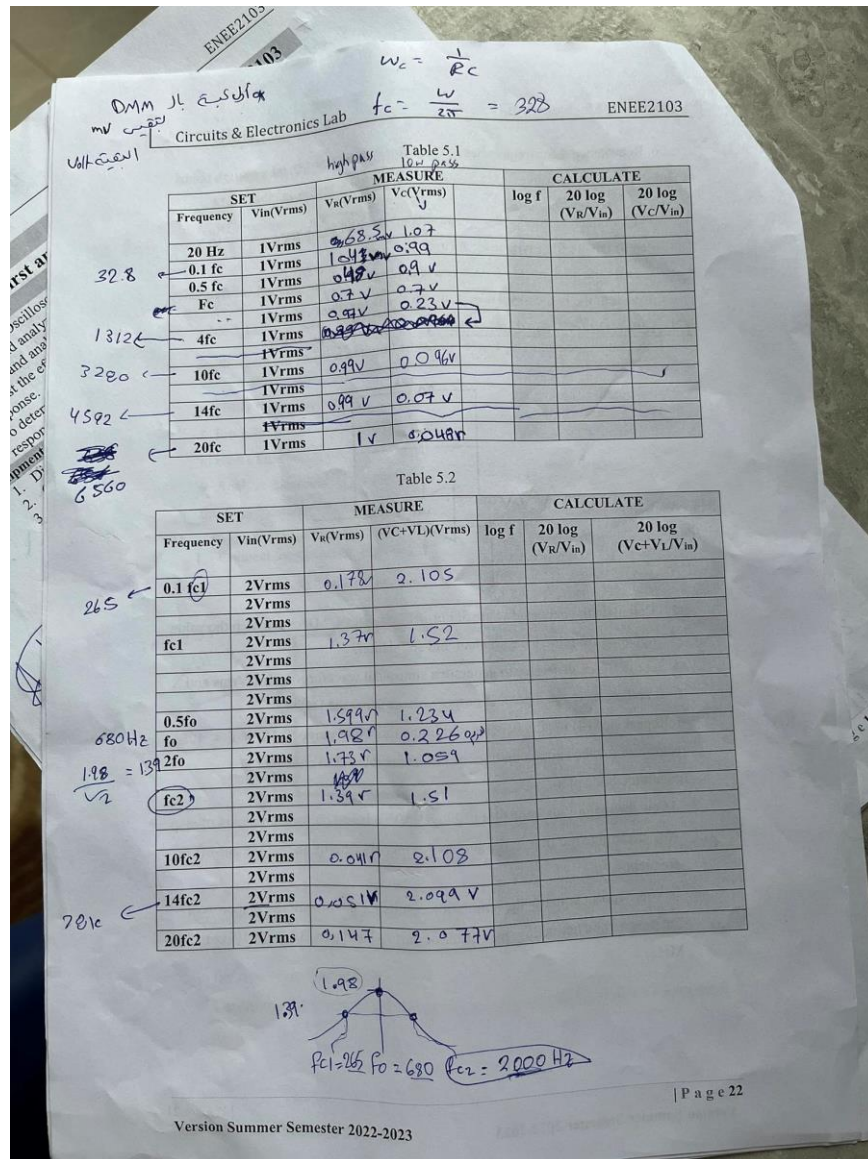
Conclusion

The experiments proved the theoretical principles and practical applications of various types of filters in electronic circuits. First and second-order passive filters were investigated along with active filters, revealing their respective frequency responses and filtering capabilities. The results revealed the design and working of filters and their importance in electronic signal processing and management.

References

- [1] "1," [Online]. Available: <https://brilliant.org/wiki/huffman-encoding/>.
- [2] "2," [Online]. Available: <https://byjus.com/jee/rl-circuit/>.
- [3] "3," [Online]. Available: <https://eepower.com/technical-articles/determining-series-rlc-circuit-resonance-frequency/#>.

Appendix



Eng.shadi hope you doing well, you forgot to assign our paper ,We were the last group who finished on Saturday and we forgot to take your assignment, please if there any problem tell us we can prove that our values are taken in the lab.